IT 5007: Software Engineering on Application Architecture

Lecture-10A

React
↓
React Native

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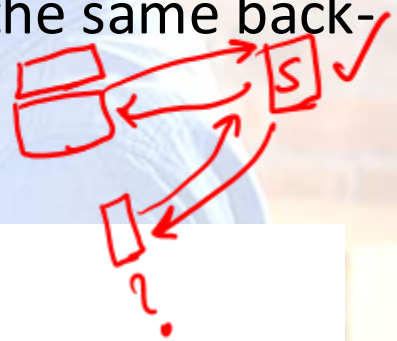
Mobile Application Development

Why do we need Mobile Applications?

- Desktop applications are obsolete.
- Web applications are still being used.
- Mobile applications are the most preferred access medium today.

- **Customer Requirements**
  - Type 1: Can you build a mobile App for xyz?
  - Type 2: We already have a web App, can you build a mobile App that works with the same back-end?



90% of all time spent on smartphones are devoted to apps

Mobile web Apps as an alternative

How is it different from developing a web application?

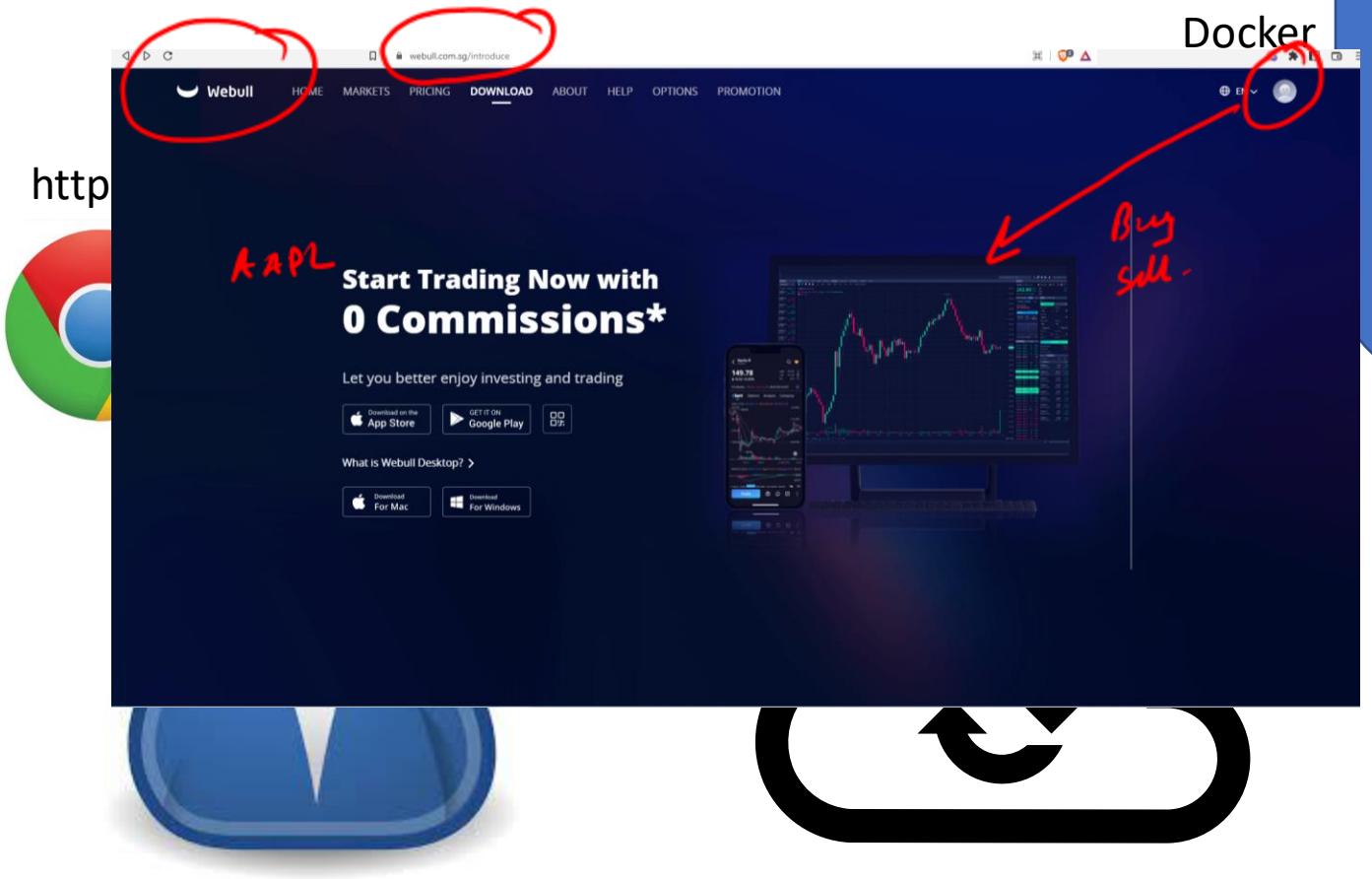
<u>Case for using Mobile Web Browser</u>

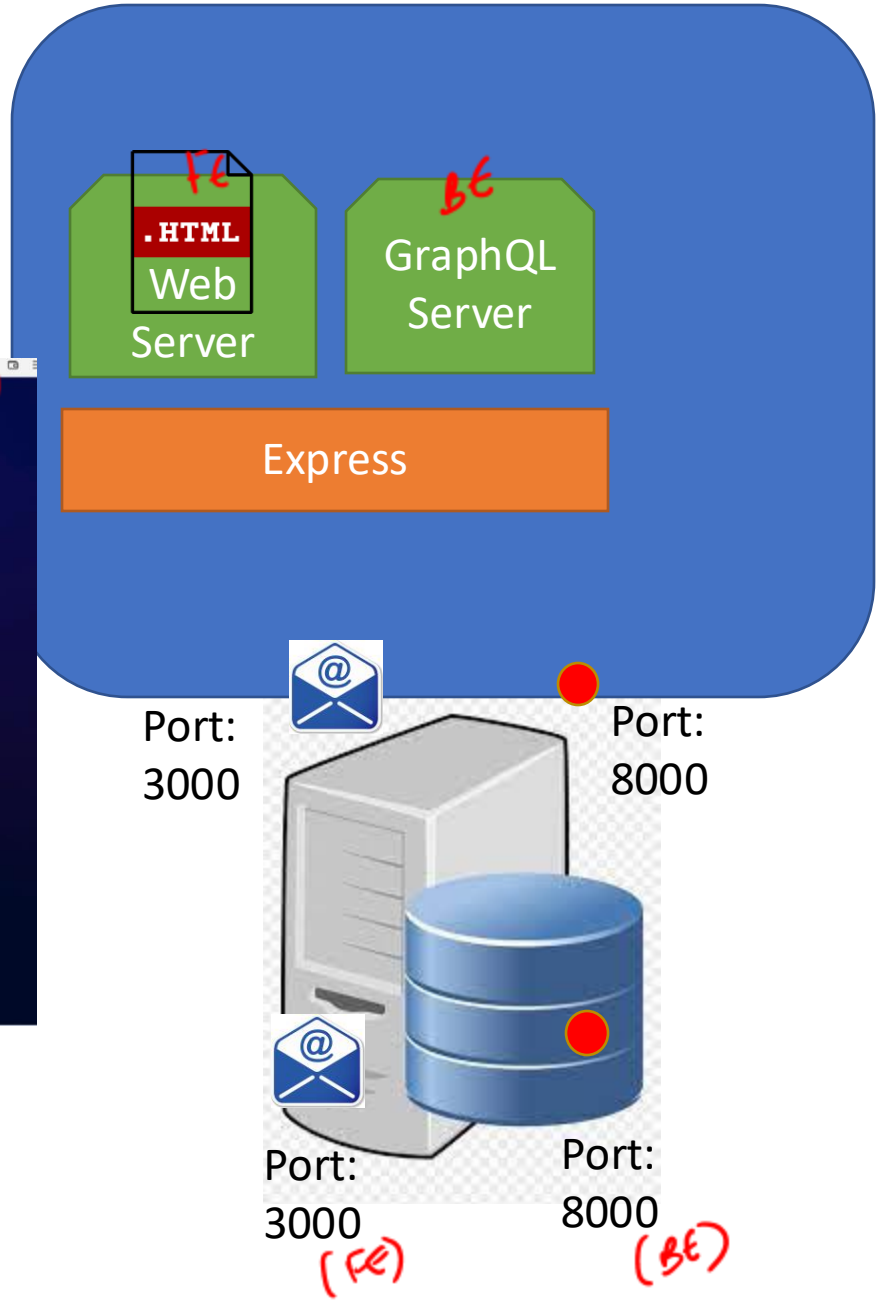
- Resize the screen
- Resize all components in the screen.
- React/JS/HTML allows you to detect the viewport of the target device (i.e., Laptop, Monitor, Mobile Device, TV) screen size.

<u>Case for a Mobile App</u>

- Design that is tailored for Mobile devices (e.g., bigger buttons).
- Touch vs. Mouse
- Downside: Android/iOS each have their own languages (Java vs. Swift) for developing Apps.

Web Application: Part-1



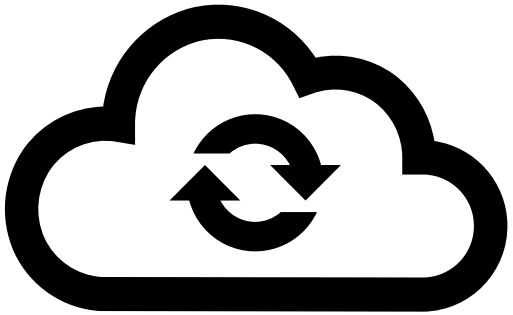


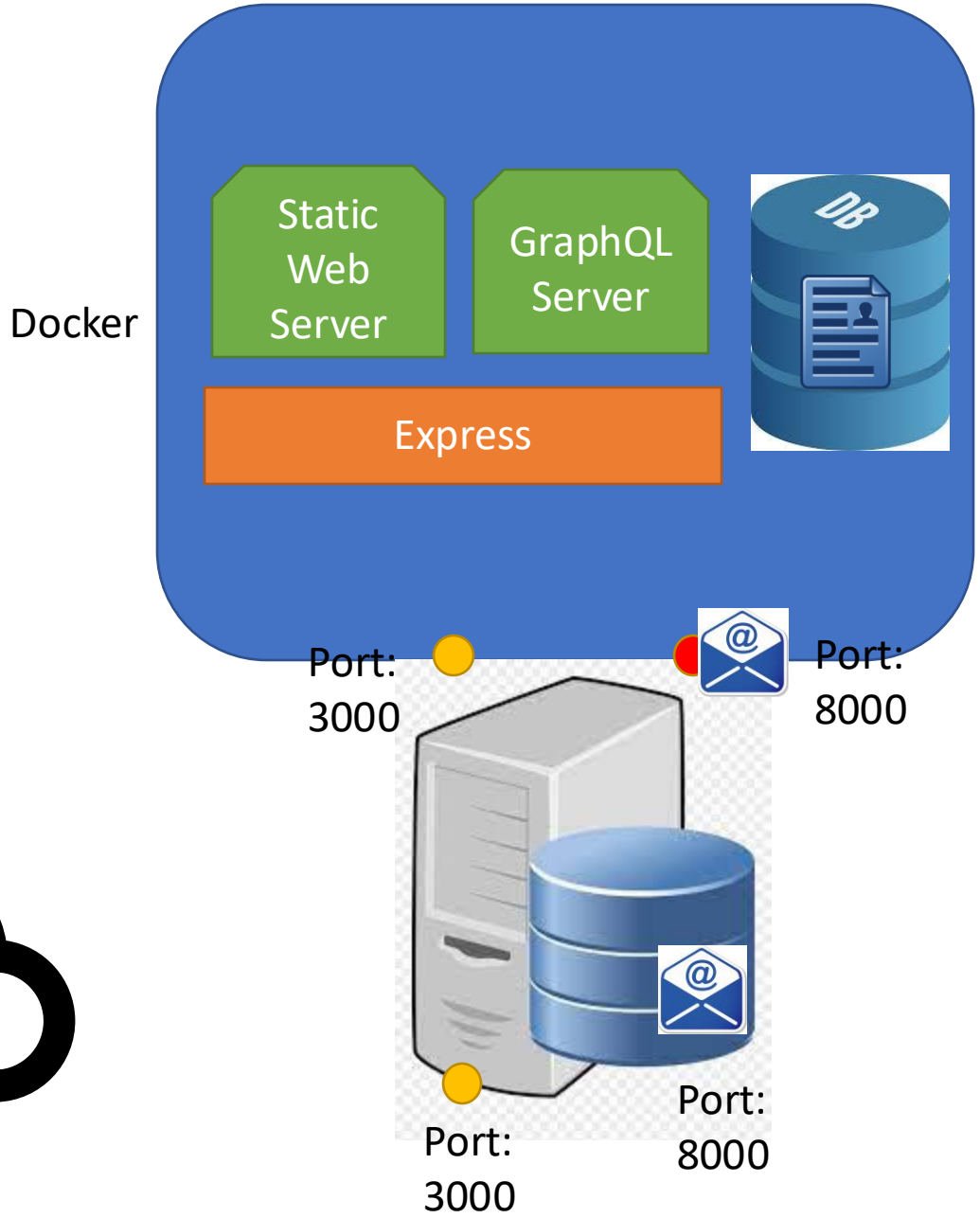
Bottomline: **Landing page** should first be delivered to the user so that user **can click some buttons and interact**.

Web Application: Part-2

http://localhost:8000/graphql

API





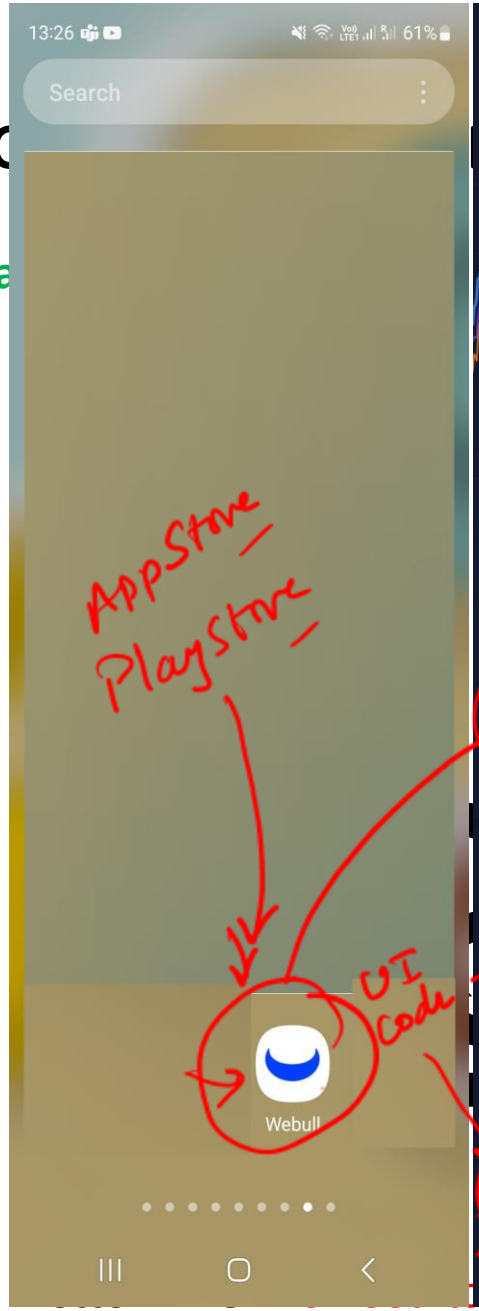
Bottomline: When the user **clicks some buttons**, API request is sent and data is received (API call).

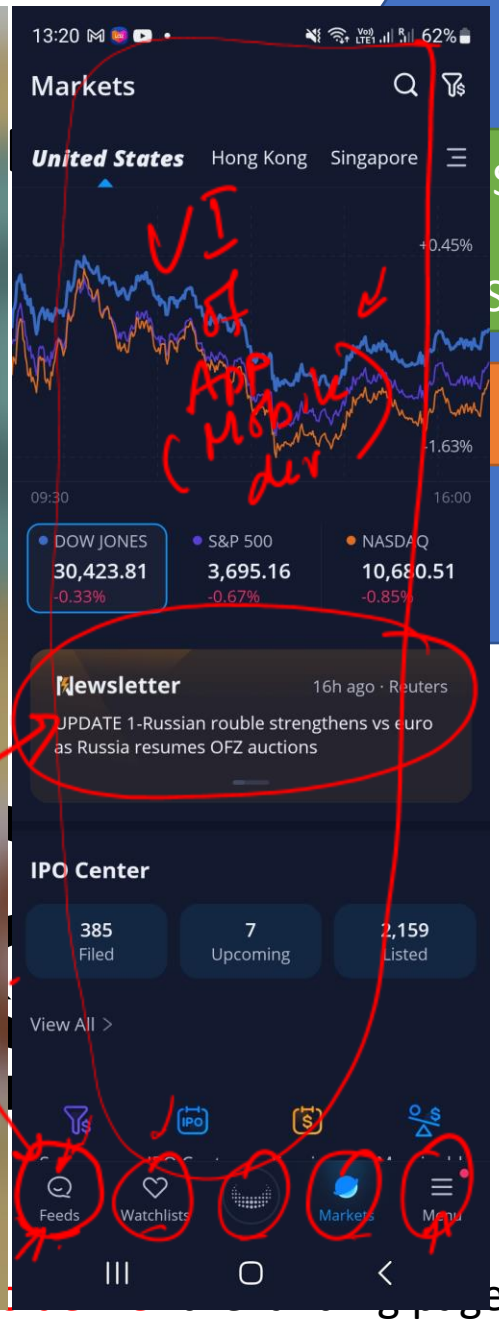
1. Data updated ✓
2. UI update
- update ✗
- SSR
- dummy elts.

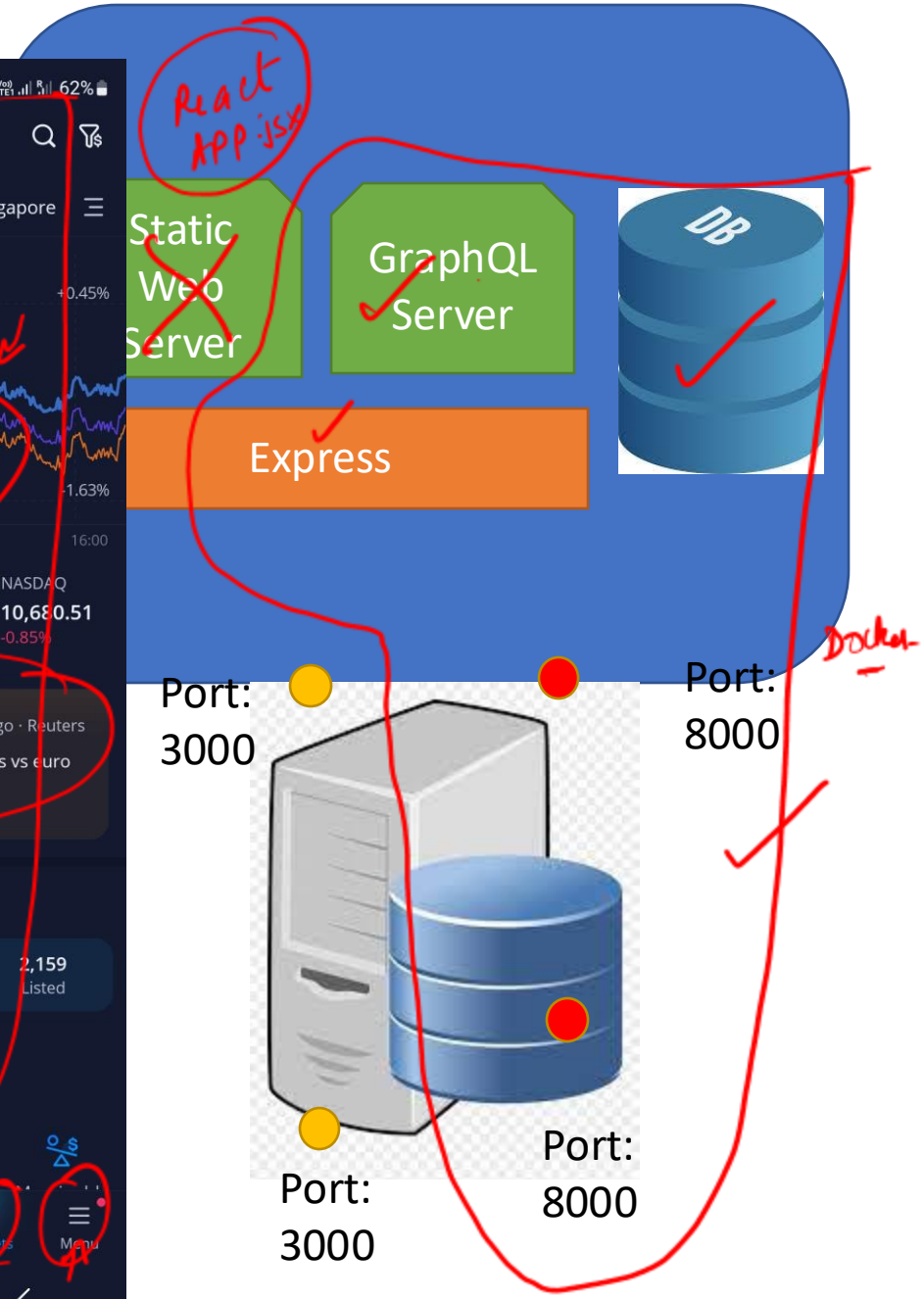
Mobile Applicatio

Premise: Landing page is alrea

App.jsx







e to the user!

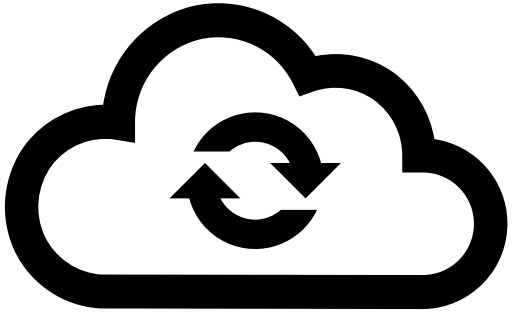
→ url → DNS → IP address

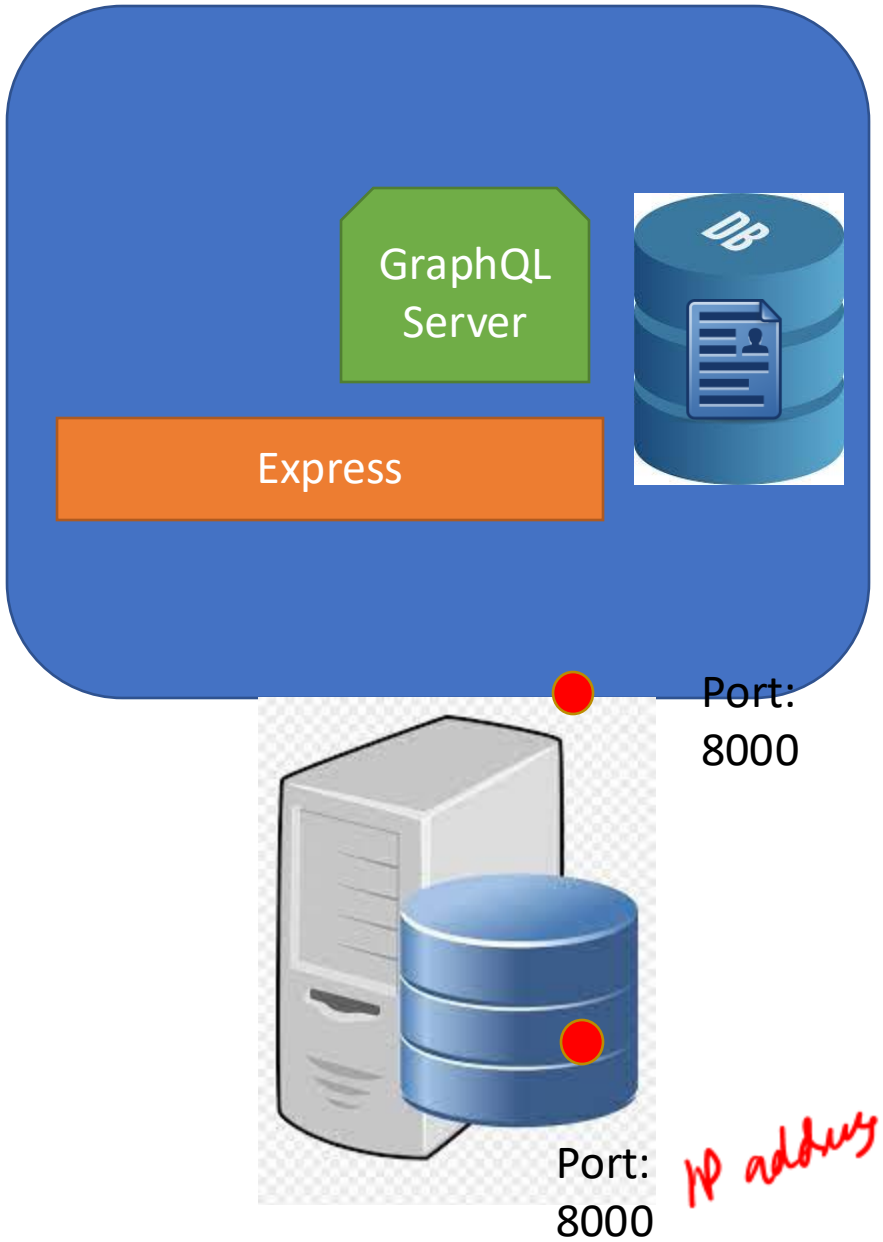
Mobile Application: How it works

Note: **Landing page/Front-end is already there**!

http://localhost:8000/graphql

→ DNS → ?x

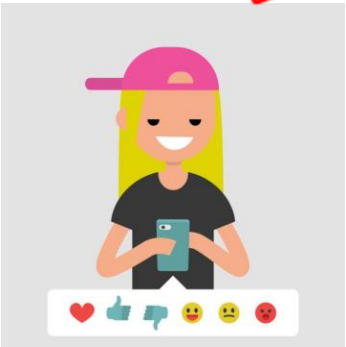




Bottomline: Only the API calls travel over the Internet each time user opens the App.

Installing a Mobile Application

How difficult can it be?

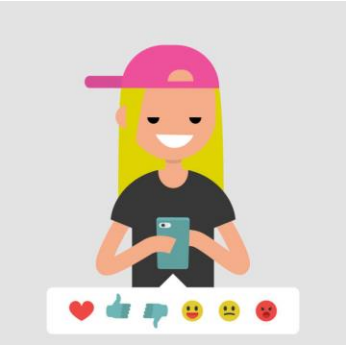


How to Install the Mobile App?

PlayStore or AppStore!

Fortnite

PDD



10% → 40%

Google play

Samsung Store

% Commission

Huawei Store

crypto

Cryptographically secure.

Challenging Process:

1. Sign the Application: https://reactnative.dev/docs/signed-apk-android
2. Publish to Play/AppStore (1.5m)

How to Install the Mobile App not on PlayStore?

Samsung (10%-20%)

apk (web)

30-40%

Commission

Legitimate

1. Sign: https://reactnative.dev/docs/signed-apk-android

Mobile App Development Setup

Wear your developer hat!

How do developers test App?

- **Production Mode**: publish to Playstore if possible (1 m)
- **Production Mode**: publish the apk to 3rd party website (Fortnite)
- **Development Mode**: No publishing. No packaging (apk). Just transfer and test it on phone.

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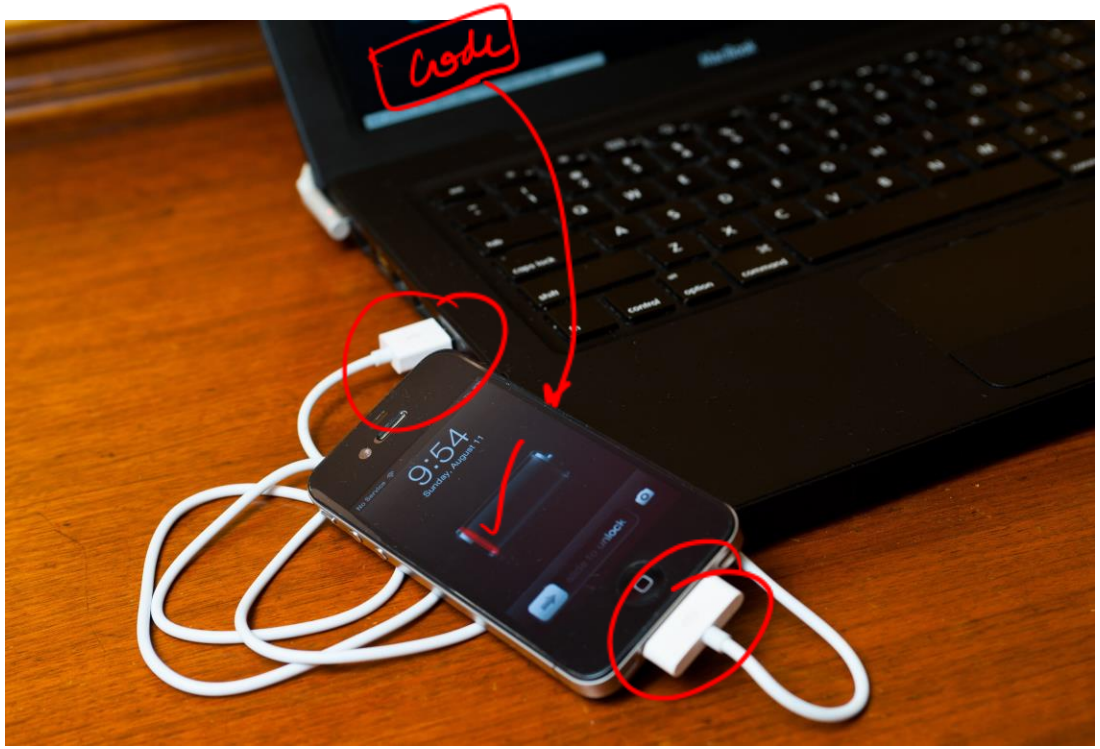
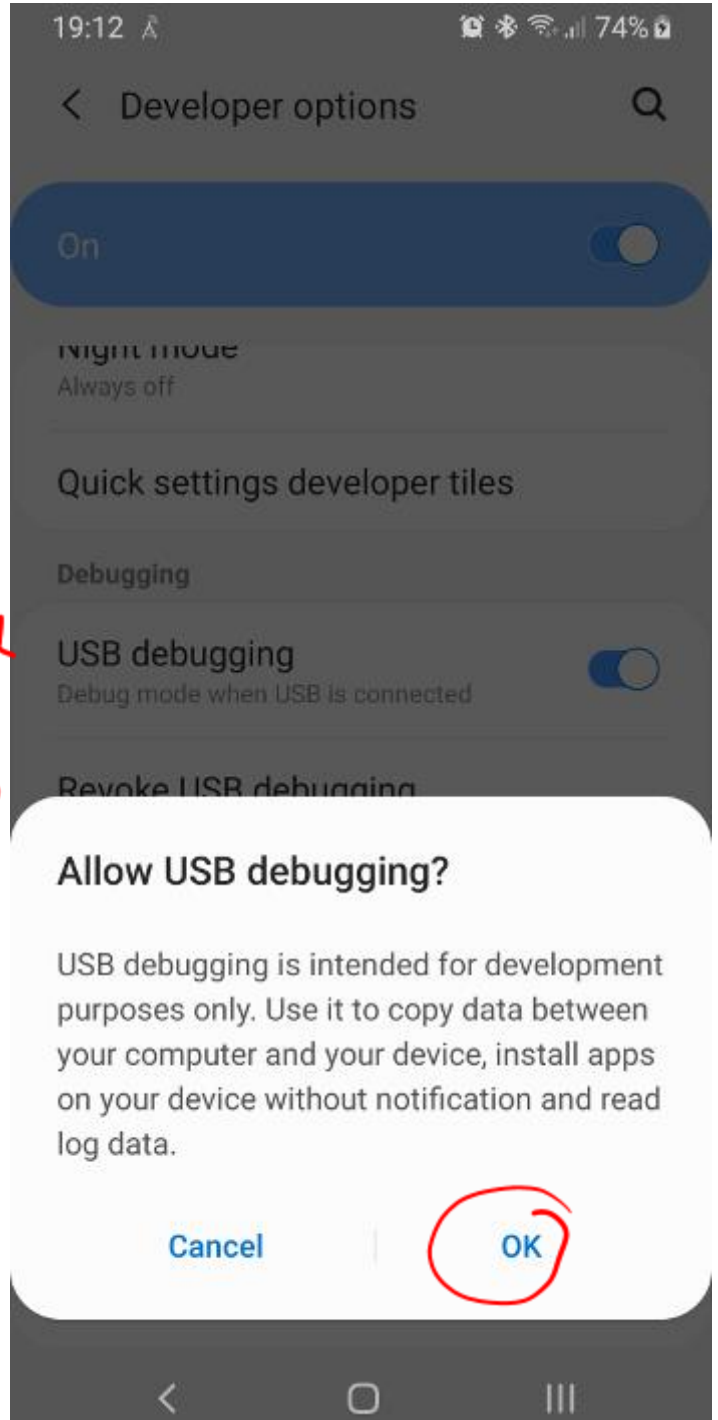
Development Mode Framework



How to Install a Mobile App that you design?

- No approval from Play~~Store~~
- No need to package/sign the app. (creator) ✓
- Isn't this a security vulnerability?



Are we done? Can we start developing?

Wait a min… Do I need an android/iPhone phone to write an App for the phone?

Emulators

My Application4 - [C:\Users\victor\AndroidStudioProjects\MyApplication4] - [app] - ...\app\src\main\java\com\example\victor\myapplication4\MainActivity.java - Android Studio 2.0 Preview

File Edit View Navigate Code Analyze Refactor Build Run Tools VCS Window Help

MyApplication4 > app > src > main > java > com > example > victor > myapplication4 > MainActivity

Android | MainActivity.java | activity_main.xml | content_main.xml | gradle.properties | app

app
Gradle Scripts
- build.gradle (Project: MyApplication4)
- build.gradle (Module: app)
- proguard-rules.pro (ProGuard Rules for app)
- gradle.properties (Project Properties)
- settings.gradle (Project Settings)
- local.properties (SDK Location)

```
import ...

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Toolbar toolbar = (Toolbar) findViewById(R.id.toolbar);
        TextView textview = (TextView) findViewById(R.id.textView);
        setSupportActionBar(toolbar);

        FloatingActionButton fab = (FloatingActionButton) findViewById(R.id.fab);
        fab.setOnClickListener((view) -> {
                Snackbar.make(view, "Replace with victor", Snackbar.LENGTH_LONG)
                        .setAction("Action", null).show();
        });

        textview.setText("hola como estan wwwwwwwwww ");

//          for (int i = 0; i < 1000000000; i++) {
//              Log.d("hola","holaaaa");
//          }

    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar if it is present.
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
```

Android Monitor

Emulator Nexus_5_API_23 Android 6.0, API 23 | com.example.victor.myapplication4 (2153)

logcat | Memory | CPU | GPU | Network

CPU | GPU

100,00 % 80,00 % 60,00 % 40,00 % 20,00 % 0,00 %

7m 30s 7m 35s 7m 40s 7m 45s 7m 50s 7m 55s

Session 'app': Launched on Nexus_5_API_23 [emulator-5554]

4: Run | TODO | 6: Android Monitor | Terminal | 0: Messages

Session 'app': Launched on Nexus_5_API_23 [emulator-5554] (9 minutes ago)

5554:Nexus_5_API_23

10:09

My Application4

Hello instantr!

hola como
estan wwww
wwwwww

virtual phone

User [0,00 %]
Kernel [0,00 %]

Activar Windows
Ir a Configuración de PC para activar Windows.

Event Log | Gradle Console

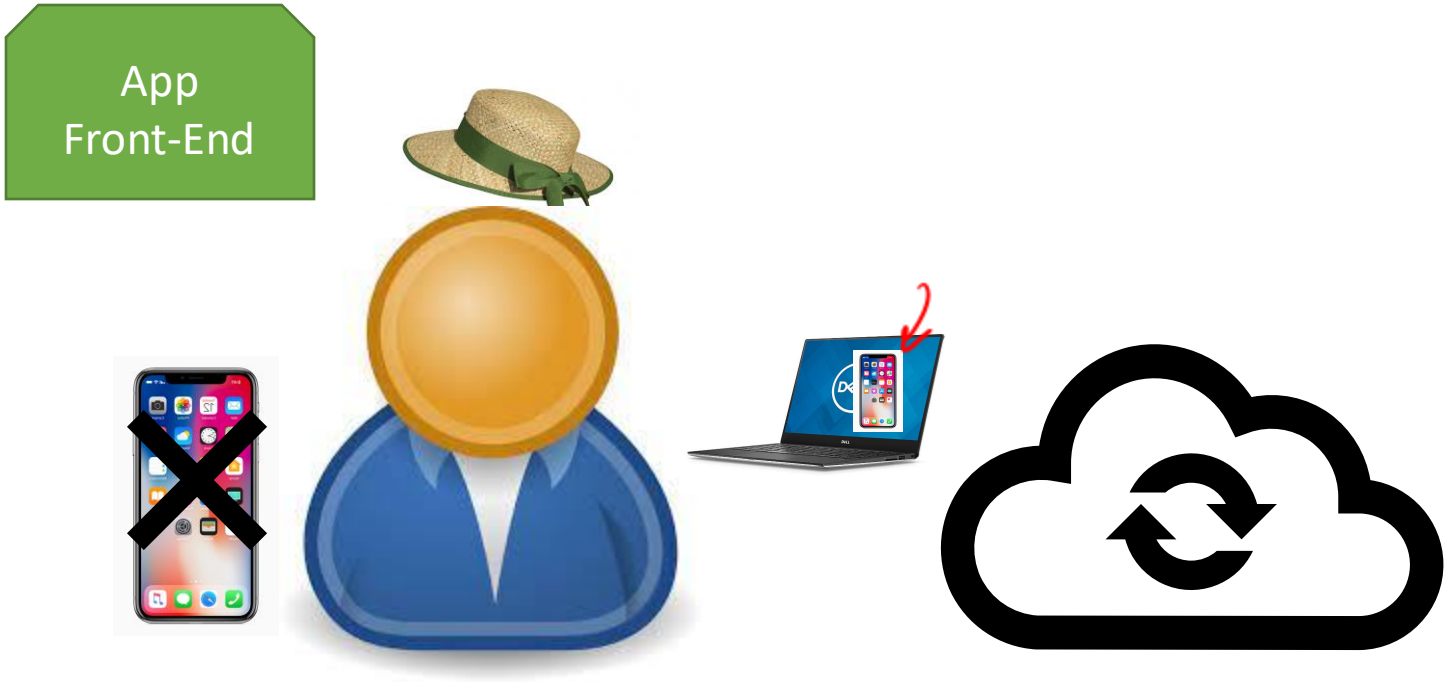
15:1 CRLF UTF-8 Context: <no context>

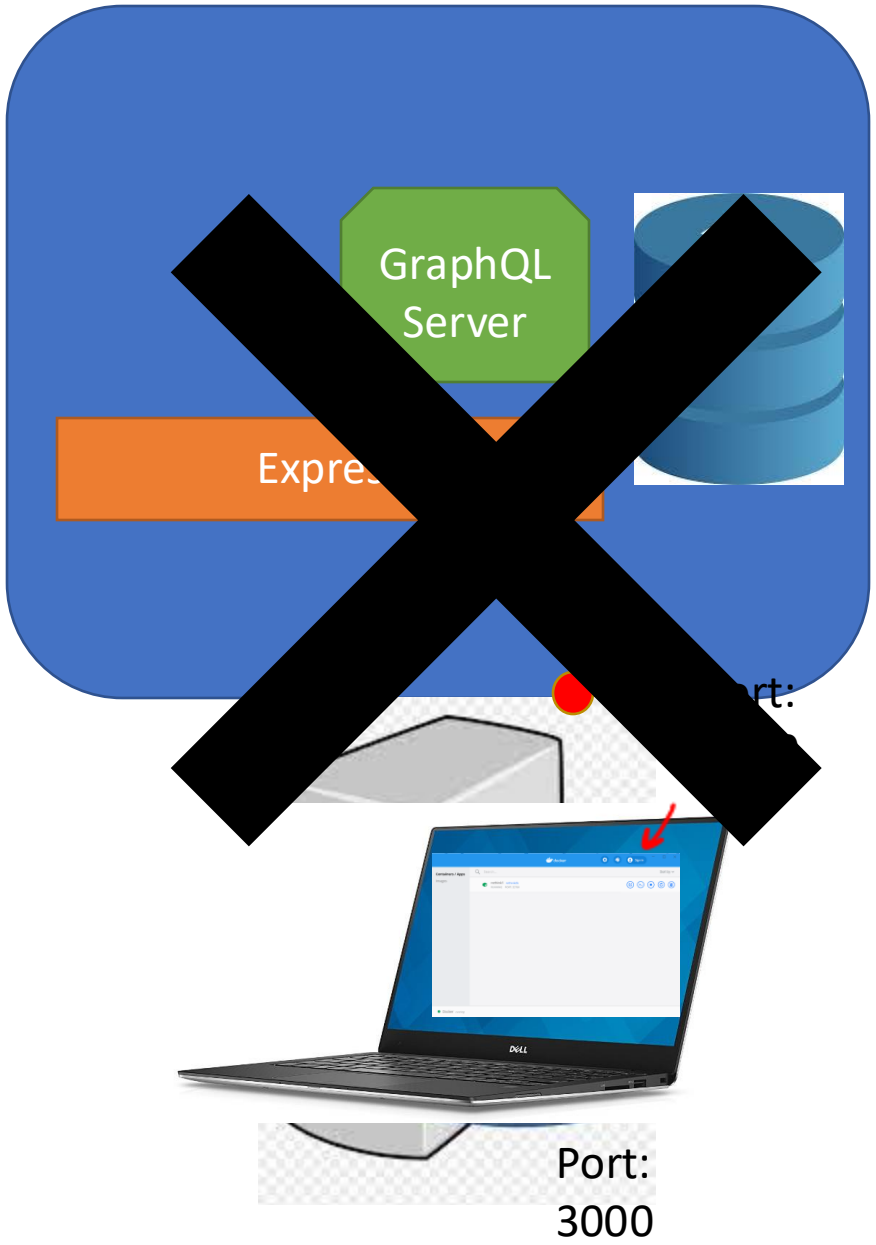
videofin... | Android ... | Desarroll... | Benzem... | H070002... | Google ... | sdf sdee... | Recibido... | Skype™ ... | android ... | ejecutar ... | Telegram | C:\xamp... | My Appl... | 5554:Ne...

18:09
10/12/2015

IT5007 Development Framework

http://localhost:3000/graphql





Codebase

Are we using the same git repository?

Comparison of Mobile vs. Web App Codebase

<u>Web App Codebase</u>

BE

- api/ directory (inside Docker)

static web

- ui/ directory (inside Docker)

<u>Mobile App Codebase</u>

- api/ directory(inside docker)
- ~~ui/ directory (inside docker)~~
- Mobile App directory (outside docker)

npx react-native init

Comparison of Mobile vs. Web App Codebase

<u>Web App Codebase</u>

- server/ directory (inside Docker)
- src and public directory (inside Docker)

<u>Mobile App Codebase</u>

- server/ directory(inside docker)
- ~~src and public directory (inside docker)~~
- Mobile App directory (outside docker, on your laptop)

ReactNative for App Development

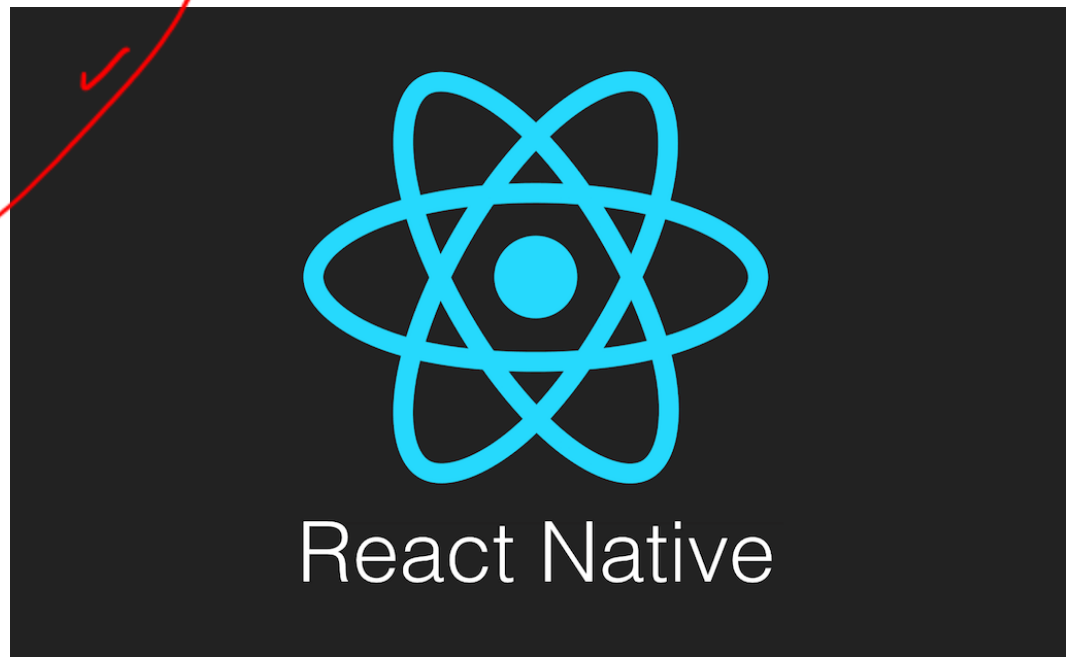
Are we developing for android or iOS?

Conventional Mobile App Dev Languages

- Android
  - Java (some C/C++)
  - Kotlin
- Blackberry
  - Java
- iOS
  - Obj C
  - Swift

Write once, Use on Android/iOS?

- ReactNative: Write code in React
- Works on both Android and iOS.
- ReactNative gets compiled to JS.
- Who compiles from JS to Java/Kotlin/Swift?



ReactNative Installation

Installation Instructions

- MacOS
- See Canvas page for instructions

- Windows
- See Canvas page for instructions
- Links to: https://github.com/pkarthik88/ReactNative

Run a Simple Application

- Create a view
- Create a label and a textbox